

Kai Barker

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EDUCATION

University of California, Santa Barbara

Santa Barbara, CA

Statistics and Data Science B.S.

Graduating June 2026

Major GPA: 3.86/4.0

Relevant Coursework: Machine Learning, Time Series, Stochastic Processes, Advanced Statistical Models, Computational Science, Differential Equations, Technology Management, Probability and Statistics, Problem Solving in Computer Science

Clubs: Consult Your Community, Statistics and Data Science Club, Data Science Collaborative, Ski and Snow Club

WORK EXPERIENCE

Consult Your Community

Santa Barbara, CA

Business Analyst

October 2025 – Present

- Produce business readiness assessment for a small business preparing for acquisition, analyzing operations, financials, and marketing to identify streamlining opportunities and reduce owner dependency
- Analyze financials, marketing strategies, and operational workflows to increase projected revenue by \$40,000
- Design 10+ Tableau visualizations and data-driven SOP to strengthen operational and financial systems

Department of Statistics and Applied Probability

Santa Barbara, CA

Undergraduate Researcher

September 2025 – Present

- Analyze 70,000+ Senate stock trades from 2012-2024 using Python, identify statistical significance in excess returns
- Calculate risk-adjusted performance using Sharpe ratios, finding members achieved 1.97 vs. benchmark of 0.79
- Develop data cleaning pipeline to validate trade reporting accuracy and filter incomplete disclosures

Foundations of Financial Technology Lab

Santa Barbara, CA

Research Assistant

September 2024 – Present

- Examined 20M+ blocks, analyzing wealth distribution across 100M+ wallets, calculating Gini coefficient of 0.96
- Produced comprehensive visualizations and statistical analysis tracking inequality evolution over a 5 year period
- Filtered institutional and Sybil addresses to reveal top 1% of wallets control 95%+ of Ether supply

Tillys

Santa Barbara, CA

Assistant Store Manager (Operations)

July 2024 – August 2025

- Directed daily operations for a high-volume retail location with \$1M+ in annual revenue, ensuring consistent performance
- Managed a team of 25+ employees, overseeing staff allocation, payroll optimization, and floor leadership to meet sales targets.
- Personally generated over \$50,000 in sales, leveraging advanced product knowledge and engagement to drive results.

PROJECTS

TrailForge AI Trail Recommendation Application

Santa Barbara, CA

SB Hacks XII - Hackathon

January 2026 – January 2026

- Built full-stack web application in 24-hours with OAuth 2.0, Python backend, and JavaScript/HTML frontend
- Engineered LLM prompts, developed responsive UI with embedded trail links, and deployed on Render
- Integrated Strava API (activity data), Google Search API (trail discovery), and Gemini API (AI recommendations)

NeurIPS Ariel Data Challenge

Santa Barbara, CA

Competition

July 2024 – September 2025

- Developed ensemble machine learning models (XGBoost + Gaussian Process Regression + Ridge) to predict exoplanet atmospheric compositions from 200+ GB of noisy spectral telescope data for ESA's Ariel mission
- Preprocessed and analyzed 14,000 complex image files across 1,100+ exoplanets, implementing advanced signal processing and noise reduction techniques for the observations
- Achieved a GLL score of 0.311 predicting atmospheric signals and uncertainties across 283 wavelength channels

NBA Player Performance Prediction

Santa Barbara, CA

Personal Project

July 2024 – Oct 2024

- Developed time series machine learning models using NBA API data to predict player performance metrics with a focus on rebound forecasting for over 450+ players.
- Engineered complex features and implemented SHAP for model interpretability, achieving a 20% accuracy improvement over baseline statistical models.
- Processed large-scale performance datasets to identify trends and validate outcomes against real-world game results.

SKILLS AND INTERESTS

Technical Skills & Languages: Python, R, SQL, JavaScript, Java, C++, CSS, HTML, Typescript, Microsoft Excel, Tableau, Feature Engineering, Statistical Modelling, Data Visualization, Data Pipeline Development, Git, API Integration, Apache Kafka, Canva, Power-Point, Pandas, NumPy, Matplotlib, Scikit-learn

Achievements: HS Valedictorian, Seal of Biliteracy (Spanish), UCSB Dean's List for Academic Excellence

Interests: Basketball, Hiking, Camping, Backpacking, Cooking, Running, Rock Climbing, Pickleball